

CYBERSECURITY CASE STUDY

EDR Alert Triage and Endpoint Investigation

A hands-on SOC investigation using endpoint telemetry, process analysis, IOCs, threat intelligence, and MITRE ATT&CK context

“I followed the evidence from the alert summary to the process chain, network activity, and threat-intelligence context before reaching a conclusion.”

Case study detail	Information
Prepared by	Osawhenomwan Mariam Iyen
Target role	Entry-Level SOC Analyst
Environment	TryHackMe simulated EDR lab
Outcome	Three detections investigated; five questions answered correctly

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Project Title

EDR Alert Triage and Endpoint Investigation

Goal / Objective

The goal of this lab was to triage endpoint detections in a simulated EDR console and use the available telemetry to understand what happened on each host. I reviewed alert summaries, process relationships, command lines, file paths, network activity, indicators of compromise, threat-intelligence labels, and MITRE ATT&CK mappings. The focus was on reaching evidence-based conclusions and documenting the findings clearly.

Lab scope: This was a controlled TryHackMe environment. Acknowledgement and manual response actions were outside the assignment scope, so I investigated and documented the detections without isolating hosts or terminating processes.

Tools Used

- TryHackMe interactive lab environment
- Simulated Endpoint Detection and Response (EDR) console
- Windows endpoint and process telemetry
- Process chain and command-line analysis
- IOC and threat-intelligence data
- MITRE ATT&CK framework
- Network activity and endpoint event evidence

What I Did

- Reviewed the EDR dashboard, identified four new detections, and prioritised the high-severity alerts first.
- Opened the Initial Access alert on DESKTOP-HR01 and reviewed the summary, risk score, process data, IOCs, and available response options.
- Traced the process chain from WINWORD.EXE to CMD.EXE, cURL.EXE, and INSTALL.EXE to confirm how the payload was downloaded.
- Recorded the payload path, malicious domain, IP address, file hash, affected user, host, and MITRE ATT&CK technique.

- Investigated the LSASS memory-access alert on WIN-ENG-LAPTOP03, confirmed the suspicious syncsvc.exe path, and reviewed the credential-dump and persistence activity.
- Analysed the network telemetry and identified the full URL used in the blocked exfiltration attempt.
- Reviewed the UpdateAgent.exe alert on DESKTOP-DEV01 and used Threat Intel context to distinguish a known internal IT utility from confirmed malicious activity.
- Documented the findings and verified all five assignment answers against the EDR evidence.

Investigation Summary

Alert 1 - Initial Access via Malicious Office Document

The first high-severity alert involved a macro-enabled document named invoice.docm on DESKTOP-HR01. The document was opened with WINWORD.EXE and triggered CMD.EXE, which then launched cURL.EXE to download install.exe. The process chain was unusual because a document application led to a command shell and a download utility.

The payload was written to C:\Users\Public\install.exe but was not executed. The IOC view linked the activity to ayeibd.thm and 1.161.138.92. The alert carried a confidence score of 95 and mapped to MITRE ATT&CK T1566.001 - Spearphishing Attachment. The combination of a macro-enabled document, suspicious child processes, an external download, and matching IOCs supported a malicious Initial Access finding.

Alert 2 - Credential Dumping via LSASS Memory Access

The second high-severity alert affected WIN-ENG-LAPTOP03. The unsigned file C:\Users\haris.khan\AppData\Local\Temp\syncsvc.exe accessed lsass.exe memory and wrote C:\Users\Public\dump_2025.dmp to disk. Threat Intel matched the executable to a known credential-dumping tool and mapped the activity to T1003.001 - LSASS Memory.

The process also showed registry activity associated with persistence. Network telemetry recorded an attempted upload to https://files-wetransfer.com/upload/session/ab12cd34ef56/dump_2025.dmp over HTTP. The destination resolved to 100.42.28.64, and the EDR blocked the attempt. The memory access, dump creation, persistence indicators, and outbound upload attempt supported a confirmed malicious credential-access finding.

Alert 3 - Execution from AppData Directory

The medium-severity alert on DESKTOP-DEV01 involved C:\Users\daniel.richards\AppData\Roaming\UpdateAgent.exe. The file was unsigned and connected to the internal IP 10.10.20.5 on port 8080 using HTTP. Those details justified investigation because AppData execution and unsigned binaries can be associated with malware.

However, Threat Intel labelled UpdateAgent.exe as a known internal IT utility tool. Based on the available evidence, I treated the event as likely legitimate but requiring normal validation, such as checking the asset owner, approved software inventory, or change record before closure. This prevented an alert from being judged on its location or unsigned status alone.

Scope note: A fourth alert, Suspicious Persistence via Scheduled Task on DESKTOP-UATSERVER, appeared in the dashboard queue. It was not part of the assignment questions and was therefore not included as a completed investigation.

Results / Findings

Detection	Host	Finding
Malicious Office Document High	DESKTOP-HR01	Malicious download chain confirmed. install.exe was written to disk but did not execute.
LSASS Memory Access High	WIN-ENG-LAPTOP03	Credential dumping and an attempted dump-file exfiltration were confirmed. The outbound attempt was blocked.
AppData Execution Medium	DESKTOP-DEV01	Suspicious execution was reviewed. Threat Intel identified UpdateAgent.exe as a known internal IT utility.

Five investigation questions were answered correctly using evidence from the EDR console. The final answers identified cURL.exe, both suspicious file paths, the complete exfiltration URL, and the Threat Intel label for UpdateAgent.exe.

Skills Demonstrated

- SOC alert triage and severity-based prioritisation
- Endpoint Detection and Response (EDR) investigation
- Windows process tree and parent-child process analysis
- Command-line, file-path, and endpoint telemetry review
- Indicator of Compromise (IOC) analysis
- Threat-intelligence validation and contextual analysis
- MITRE ATT&CK mapping: T1566.001 and T1003.001
- Credential-dumping and data-exfiltration investigation
- Network activity and URL analysis
- False-positive assessment and evidence-based decision making
- Incident documentation and clear technical reporting

What I Learned

This investigation taught me that an EDR alert is the starting point, not the final answer. The alert title and severity helped me decide where to begin, but the process chain, command line, IOCs, network activity, and Threat Intel provided the evidence needed to understand what actually happened.

I also learned why context matters during triage. UpdateAgent.exe looked suspicious because it was unsigned, ran from AppData, and made an outbound connection. Threat Intel showed that it was a known internal utility, so I would validate it before taking disruptive action. This improved my understanding of false positives and the importance of protecting business operations while investigating security alerts.

Finally, the lab strengthened my ability to follow an attack across different types of telemetry and record exact evidence. I am more confident reading process relationships, identifying malicious behaviour, checking IOCs, mapping activity to MITRE ATT&CK, and explaining my reasoning in a way that another analyst can follow.

Portfolio keywords: EDR | Alert Triage | SOC Investigation | Process Tree Analysis | Windows Telemetry | IOC Analysis | Threat Intelligence | MITRE ATT&CK | Credential Access | Network Analysis | Incident Documentation

Investigation Evidence Walkthrough

The screenshots below are arranged in the same order as the investigation. They show the alert queue, the evidence reviewed in each EDR tab, and the final assignment results.

1. Starting the Lab and Prioritising Alerts

Evidence 1: Lab scenario and investigation questions

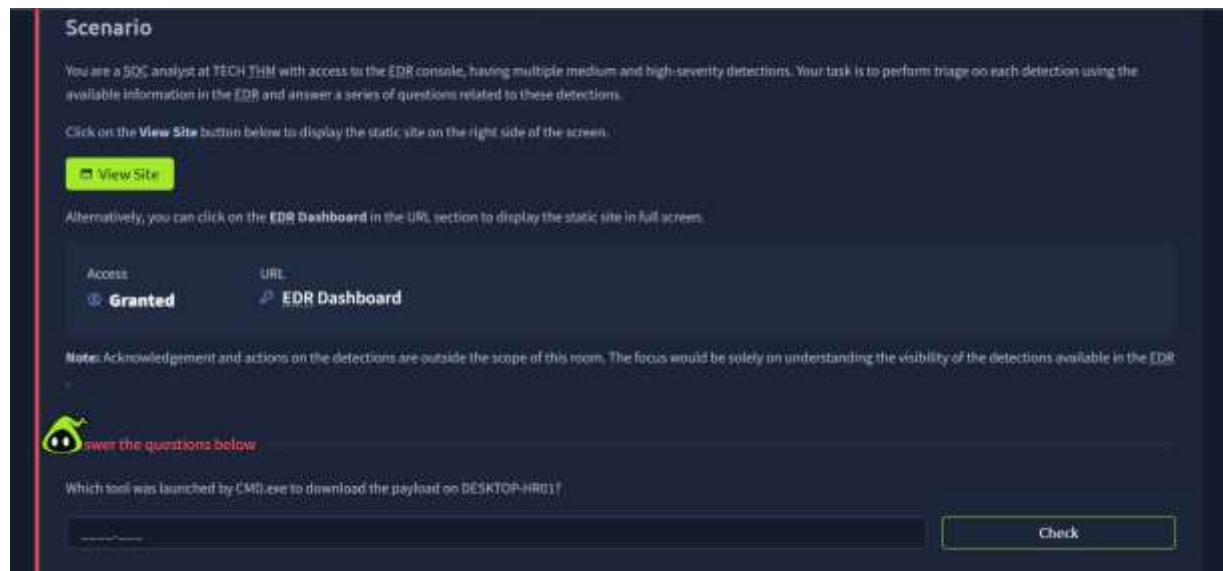


Figure 1. Lab scenario and investigation questions

Why it mattered: The task required triage of several medium- and high-severity EDR detections using the information available in the simulated console.

Evidence 2: EDR dashboard and recent detection queue

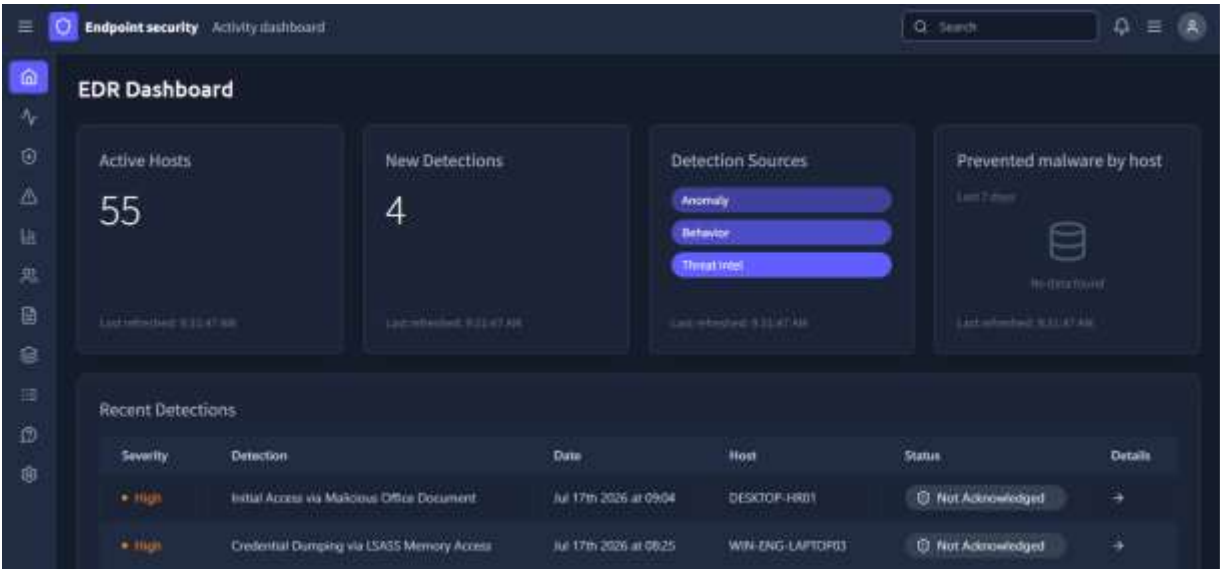


Figure 2. EDR dashboard and recent detection queue

Why it mattered: The dashboard showed 55 active hosts and four new detections. I began with the high-severity Initial Access alert on DESKTOP-HR01.

Evidence 3: Remaining alerts and MITRE tactic overview



Figure 3. Remaining alerts and MITRE tactic overview

Why it mattered: The queue also contained credential dumping, AppData execution, and scheduled-task persistence alerts. The chart provided a wider view of detection tactics across the environment.

2. Investigating the Malicious Office Document

Evidence 4: Initial Access alert summary

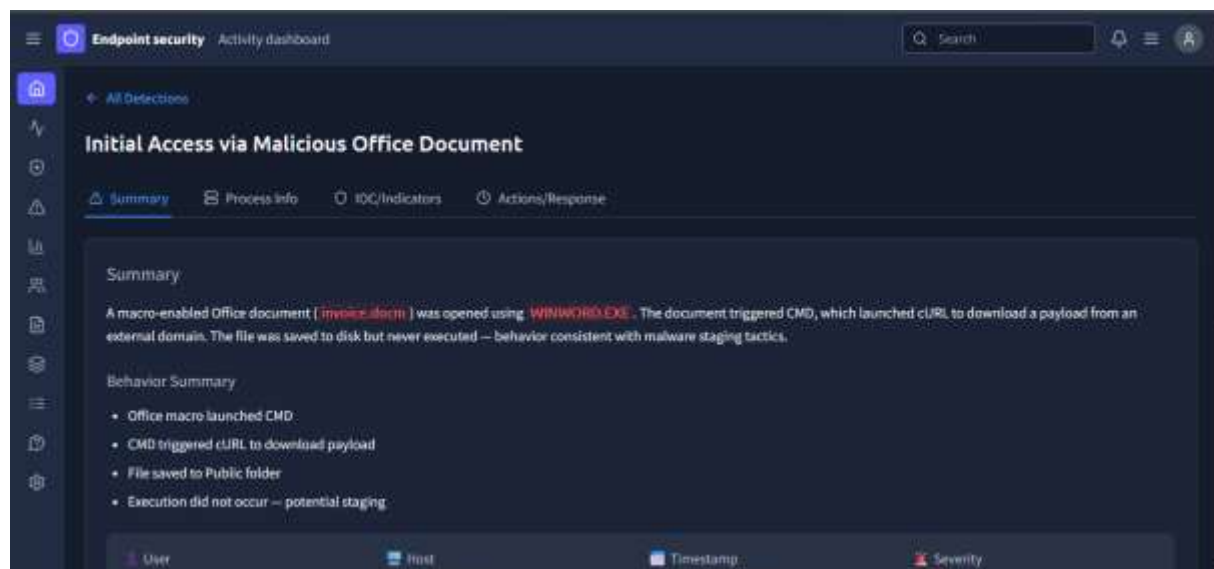


Figure 4. Initial Access alert summary

Why it mattered: The summary connected invoice.docm, WINWORD.EXE, CMD, cURL, and a downloaded payload. It also noted that the payload was saved but not executed.

Evidence 5: Initial Access alert details and risk assessment

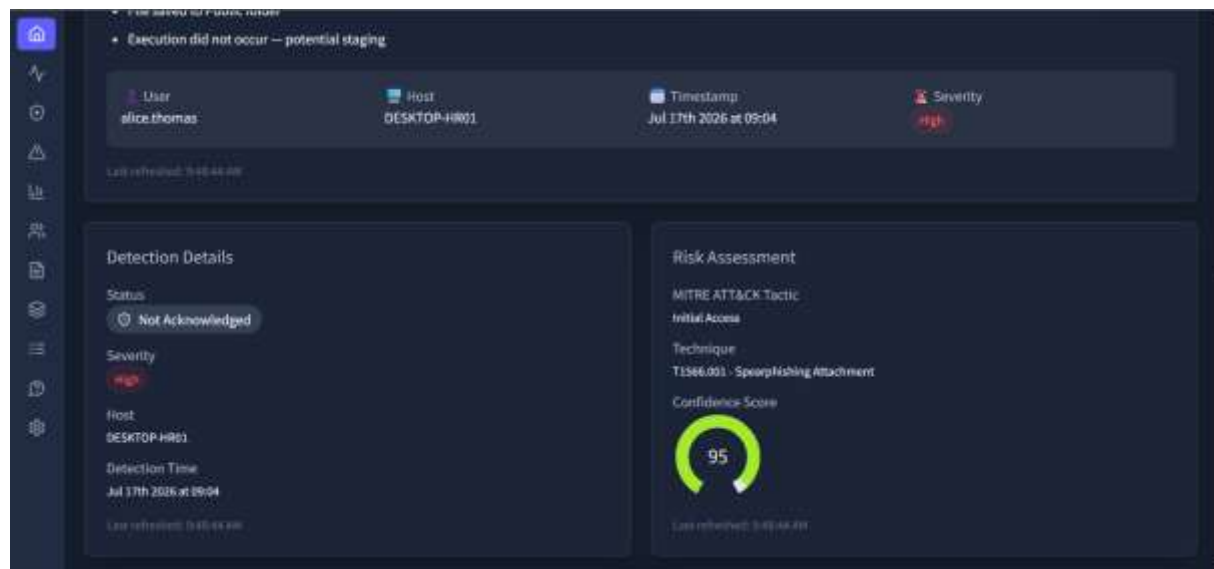


Figure 5. Initial Access alert details and risk assessment

Why it mattered: The alert affected DESKTOP-HR01, was rated High severity, carried a confidence score of 95, and mapped to T1566.001 - Spearfishing Attachment.

Evidence 6: Process chain review: Microsoft Word

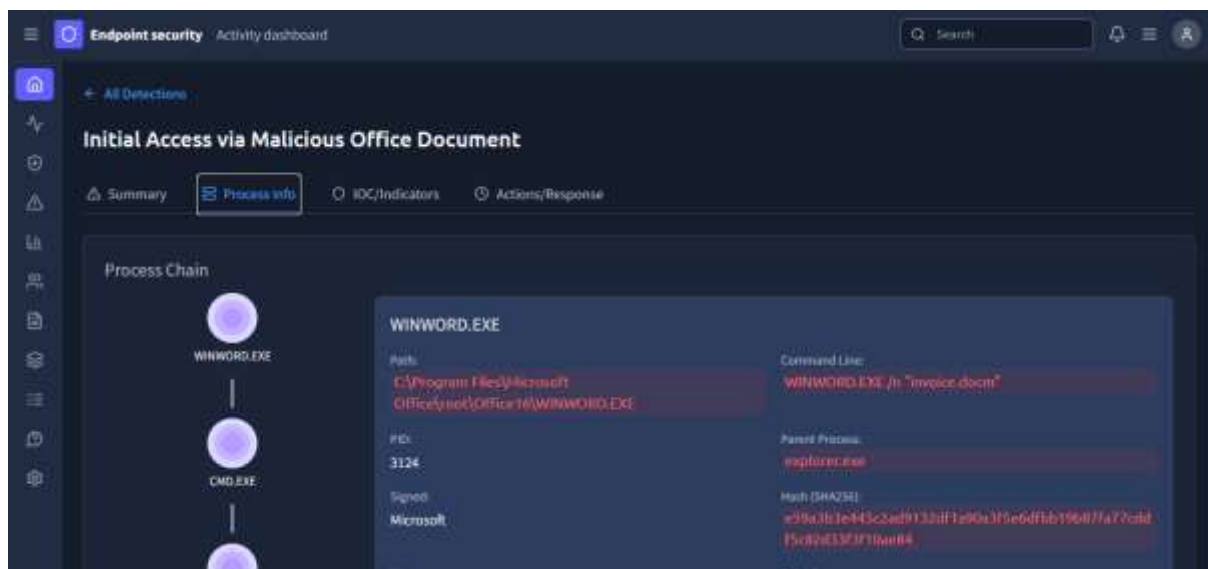


Figure 6. Process chain review: Microsoft Word

Why it mattered: I opened Process Info to confirm that WINWORD.EXE launched after the user opened the macro-enabled document.

Evidence 7: Process chain review: CMD, cURL, and payload

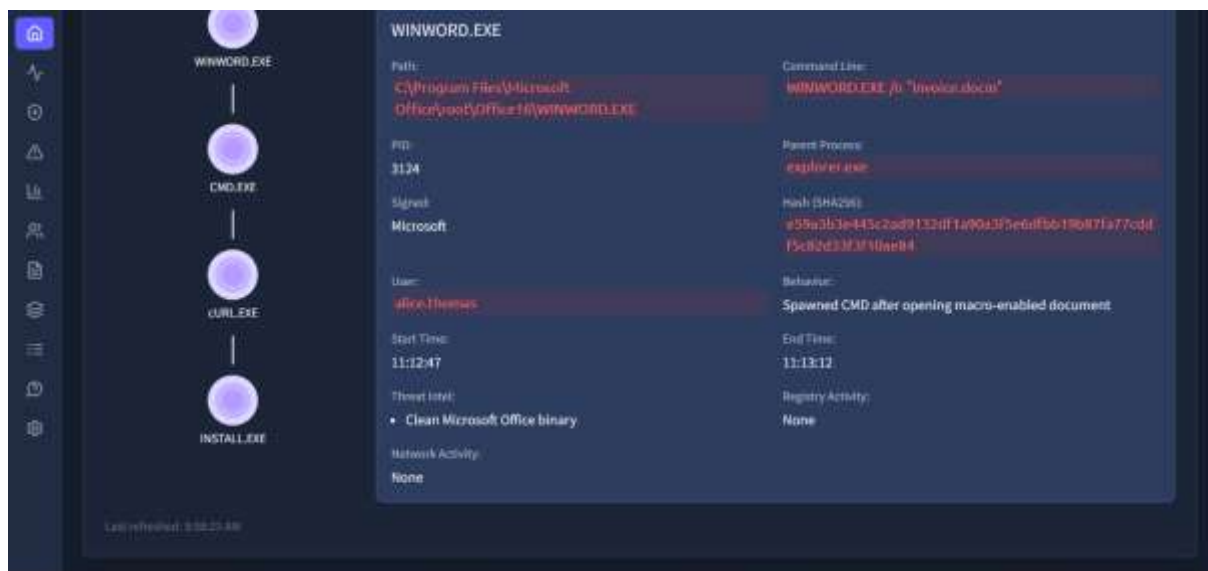


Figure 7. Process chain review: CMD, cURL, and payload

Why it mattered: The full process chain showed WINWORD.EXE -> CMD.EXE -> cURL.EXE -> INSTALL.EXE, confirming the tool used to download the payload.

Evidence 8: Initial Access indicators of compromise

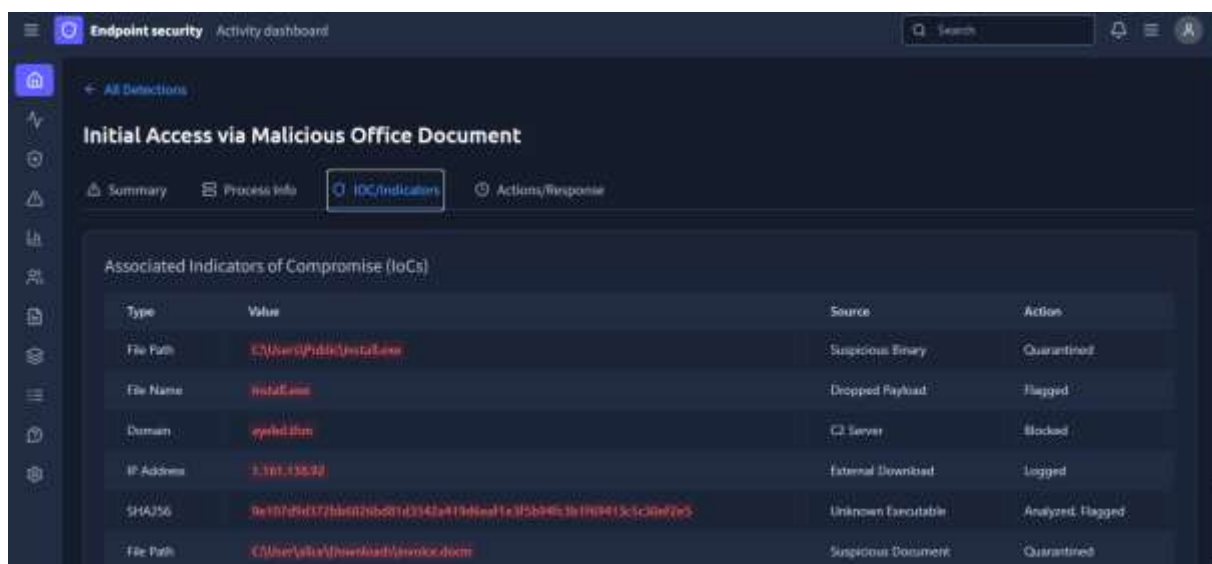


Figure 8. Initial Access indicators of compromise

Why it mattered: The IOC view recorded the payload path, file name, domain, IP address, SHA256 value, and original suspicious document path.

Evidence 9: Available response options

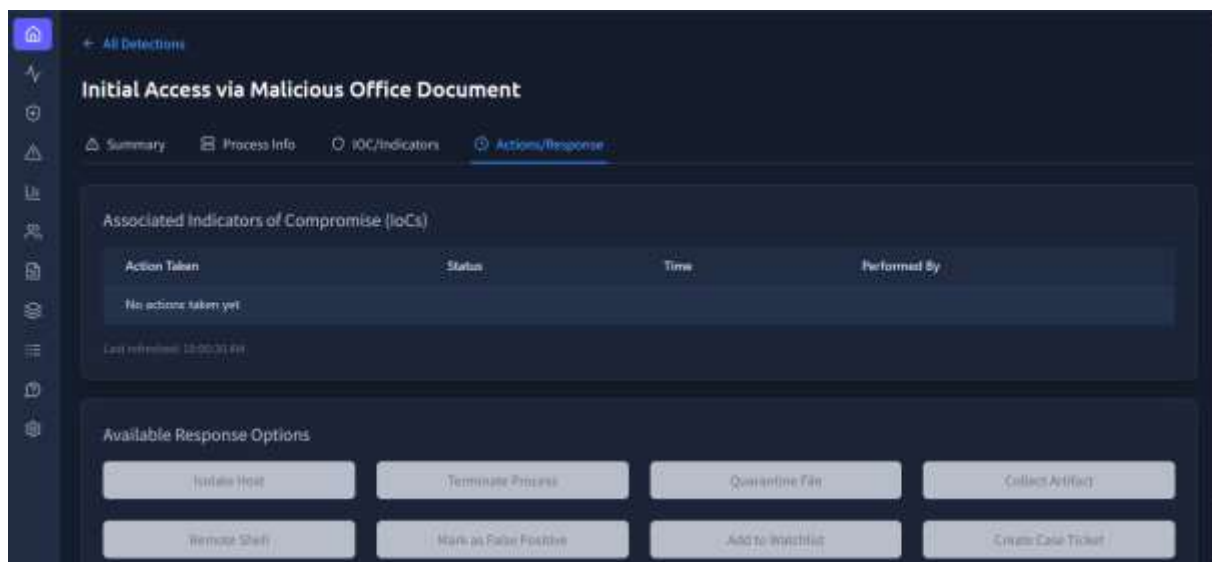
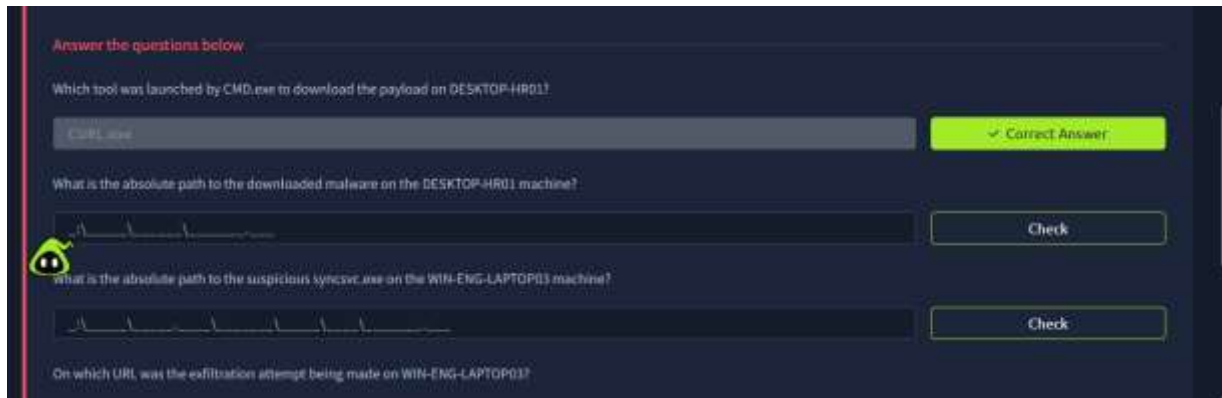


Figure 9. Available response options

Why it mattered: The console displayed containment and investigation options, but no manual actions had been taken. The lab instructions limited the task to detection visibility and triage.

Evidence 10: Verification of the first finding



The screenshot shows a quiz interface with a dark blue background. At the top, it says "Answer the questions below". The first question is "Which tool was launched by CMD.exe to download the payload on DESKTOP-HR01?". The answer field contains "cURL.exe" and a green button labeled "✓ Correct Answer!". The second question is "What is the absolute path to the downloaded malware on the DESKTOP-HR01 machine?". The answer field is empty, and there is a green "Check" button. The third question is "What is the absolute path to the suspicious syncsvc.exe on the WIN-ENG-LAPTOP03 machine?". The answer field is empty, and there is a green "Check" button. The fourth question is "On which URL was the exfiltration attempt being made on WIN-ENG-LAPTOP03?". The answer field is empty. A small green alien icon is visible on the left side of the interface.

Figure 10. Verification of the first finding

Why it mattered: The lab accepted cURL.exe as the tool launched by CMD.exe. I then used the IOC data to identify the downloaded malware path.

3. Investigating LSASS Credential Dumping

Evidence 11: Credential-dumping indicators

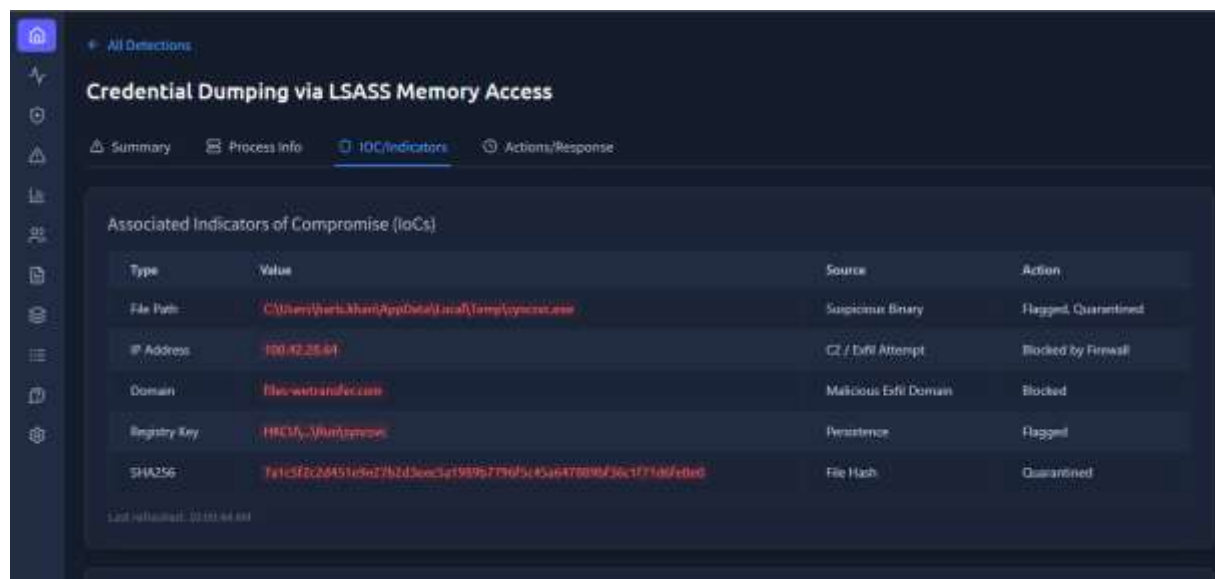


Figure 11. Credential-dumping indicators

Why it mattered: The second investigation identified syncsvc.exe in a user's Temp directory, a suspicious exfiltration domain and IP, persistence-related registry activity, and a quarantined file hash.

Evidence 12: Verification of file-path findings

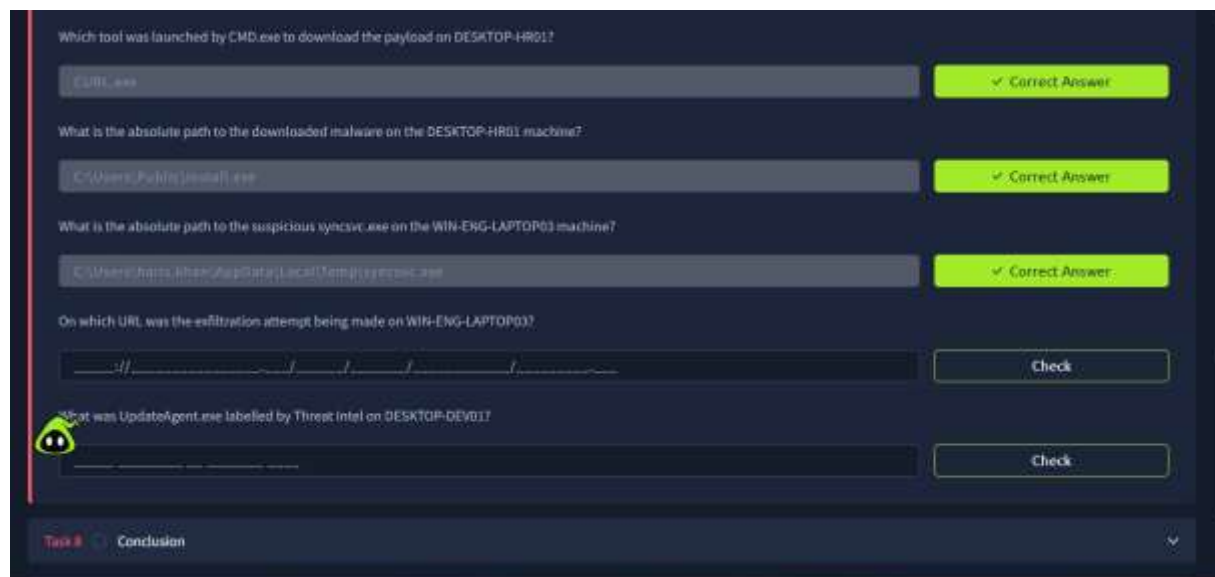


Figure 12. Verification of file-path findings

Why it mattered: The first three questions were confirmed, including the paths to install.exe and syncsvc.exe. The remaining work was to identify the full exfiltration URL and validate UpdateAgent.exe.

Evidence 13: syncsvc.exe process analysis



Figure 13. syncsvc.exe process analysis

Why it mattered: Process Info showed an unsigned executable that accessed lsass.exe memory and wrote C:\Users\Public\dump_2025.dmp to disk.

Evidence 14: Blocked credential-dump exfiltration attempt

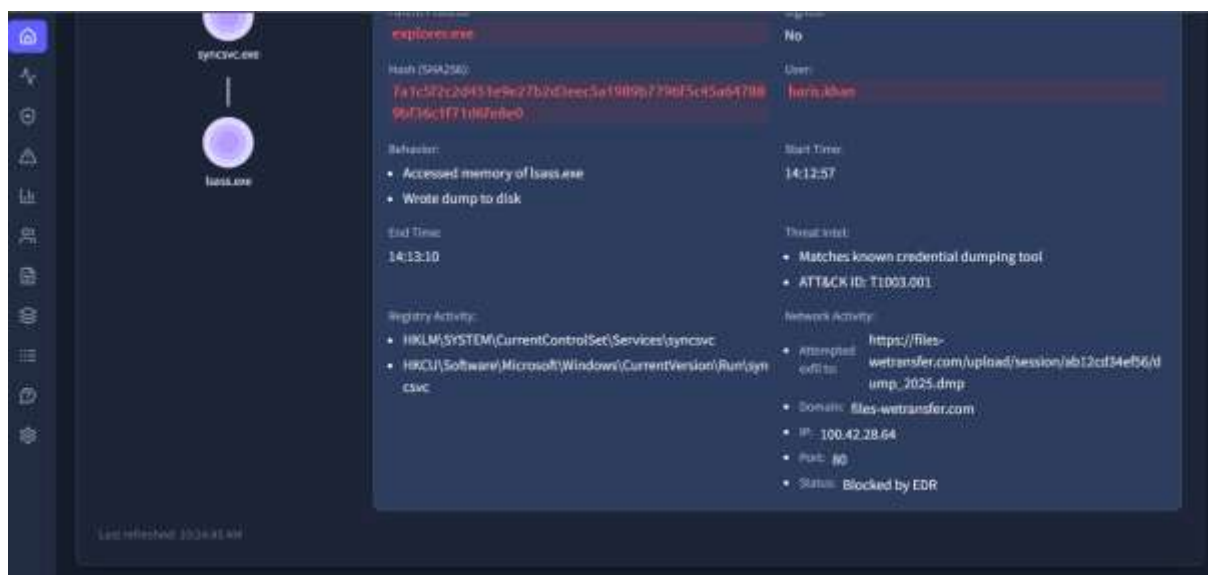


Figure 14. Blocked credential-dump exfiltration attempt

Why it mattered: Network Activity showed an HTTP upload attempt to files-wetransfer.com on port 80. The EDR blocked the connection before the dump file could leave the environment.

Evidence 15: Exact exfiltration URL captured from telemetry

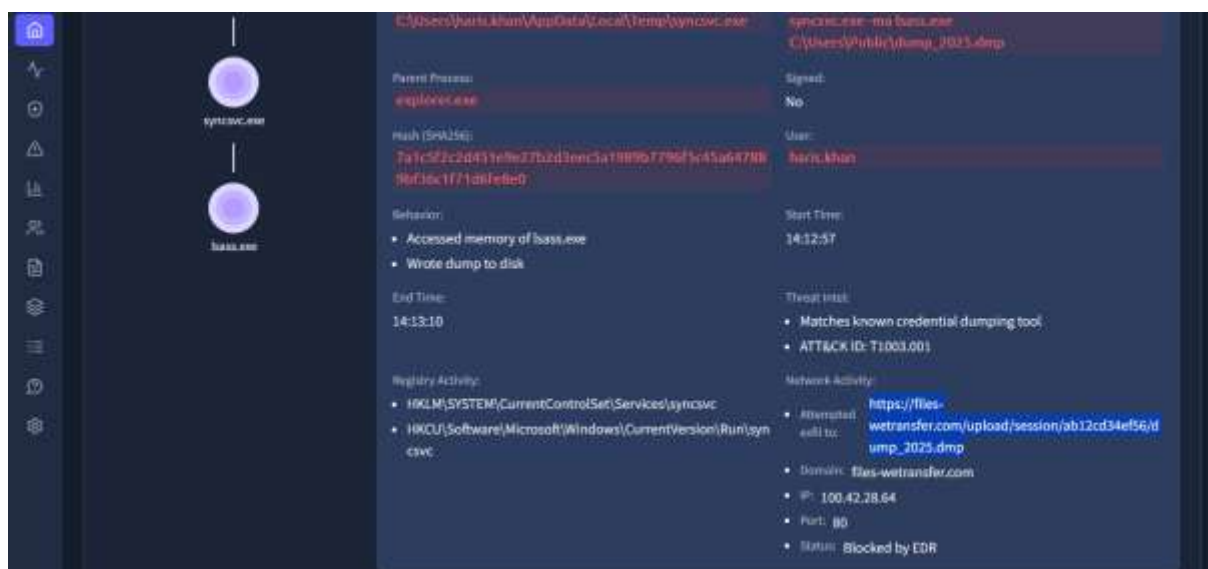


Figure 15. Exact exfiltration URL captured from telemetry

Why it mattered: I copied the complete URL from the Network Activity field instead of relying only on the domain IOC.

Evidence 16: Verification of the exfiltration URL

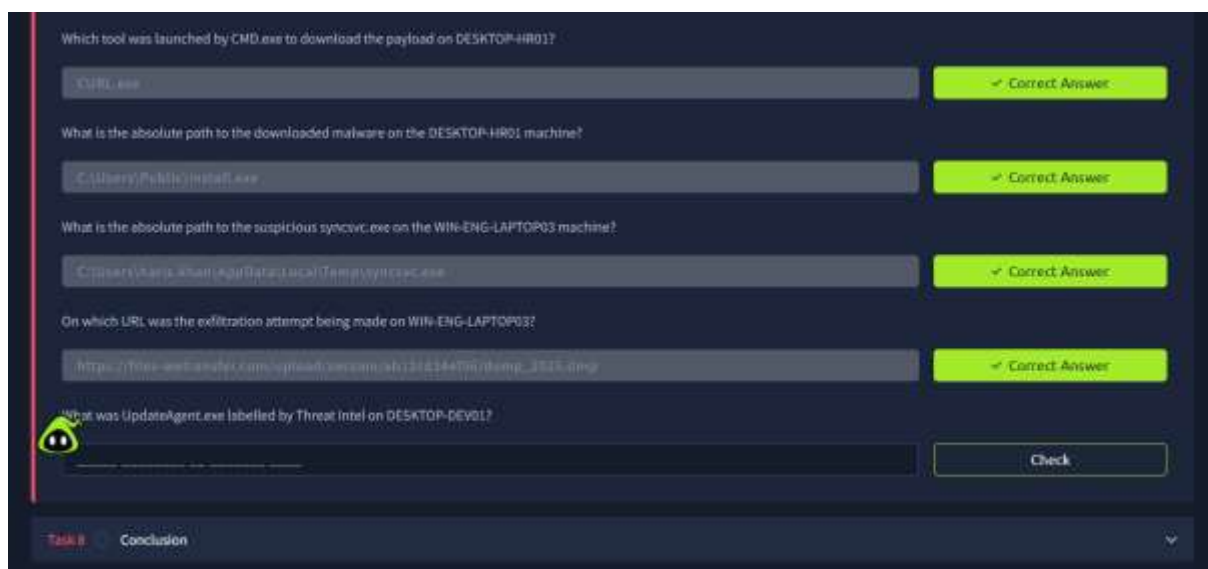


Figure 16. Verification of the exfiltration URL

Why it mattered: The full URL was accepted, leaving the AppData execution alert as the final investigation task.

4. Validating the AppData Execution Alert

Evidence 17: UpdateAgent.exe process review

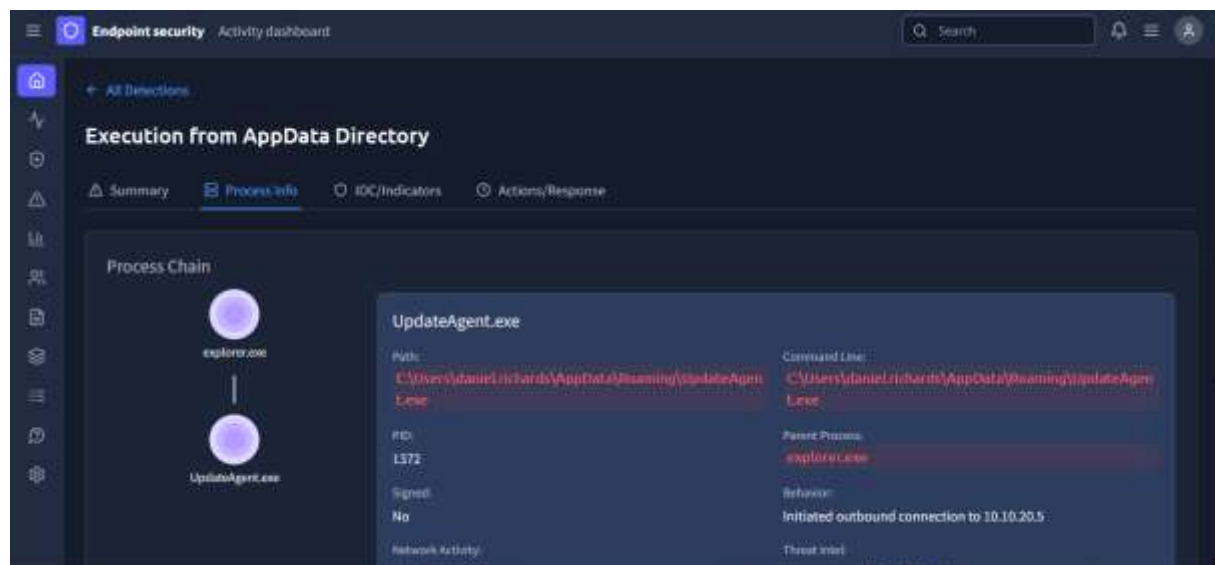


Figure 17. UpdateAgent.exe process review

Why it mattered: The executable ran from the user's AppData\Roaming folder, was unsigned, and initiated an outbound connection, which explained why the EDR generated a medium-severity alert.

Evidence 18: Threat intelligence context for UpdateAgent.exe

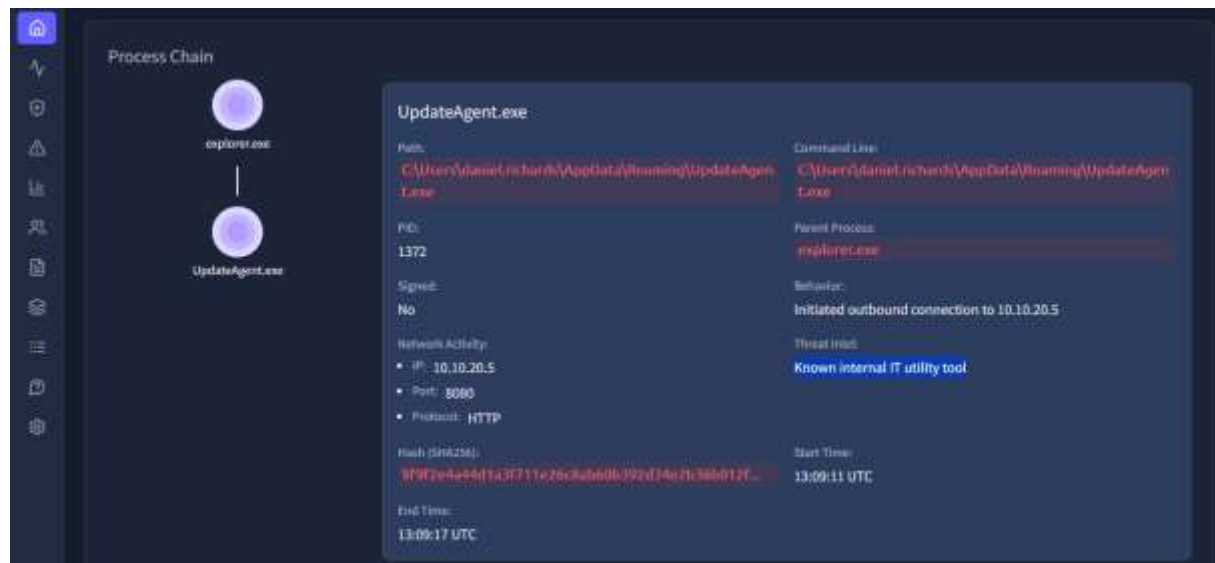
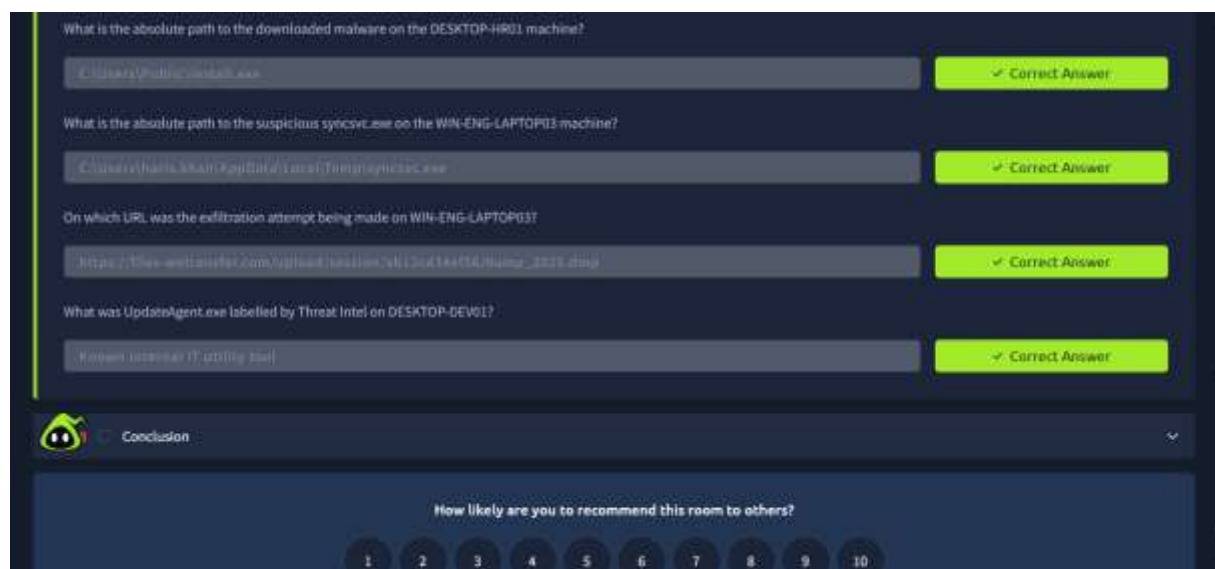


Figure 18. Threat intelligence context for UpdateAgent.exe

Why it mattered: Threat Intel labelled the file as a known internal IT utility tool. This context reduced the likelihood that the activity was malicious, although normal validation would still be required before closure.

Evidence 19: Completed lab results



The screenshot displays a dark-themed interface for a lab completion. It contains four questions, each with a text input field showing the correct answer and a green button labeled 'Correct Answer'.

- Question 1: "What is the absolute path to the downloaded malware on the DESKTOP-HR01 machine?"
Answer: `C:\Users\Public\Downloads.exe`
- Question 2: "What is the absolute path to the suspicious syncsvt.exe on the WIN-ENG-LAPTOP03 machine?"
Answer: `C:\Users\Harris\AppData\Local\Temp\syncsvt.exe`
- Question 3: "On which URL was the exfiltration attempt being made on WIN-ENG-LAPTOP03?"
Answer: `https://files-website8f8.com/vgloak/tacllaxvN13ut1a4f16/0umz_3278.dmg`
- Question 4: "What was UpdateAgent.exe labelled by Threat Intel on DESKTOP-DEV01?"
Answer: `Known: UACB6437T: Utility [low]`

Below the questions is a section titled "Conclusion" with a small icon. At the bottom, there is a question: "How likely are you to recommend this room to others?" followed by a row of ten numbered buttons (1-10).

Figure 19. Completed lab results

Why it mattered: All five investigation questions were answered correctly, confirming the process, path, URL, and threat-intelligence findings recorded in the case study.

Closing Note

This case study reflects a simulated SOC investigation completed for hands-on learning. The evidence, decisions, and conclusions are based on the telemetry provided in the lab. No production systems or real user data were accessed.